

Arya Darshan Roi Vijai

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ACADEMIC QUALIFICATION

Masters in Urban Data Science New York University, New York City; Expected May'27

Bachelor of Technology in Electronics and Communications Engineering PES University, Bengaluru; CGPA: 6.79; Oct'23

WORK EXPERIENCE

Graduate Research Student, New York University, New York City

Sep'23-present

- Use of LLMs for image-text generation from satellite image data of wildfire prone regions
- Analysis of LLMs for urban wildfire recovery

Software Developer/Engineer, SIEMENS, Bengaluru

Jun'23-Aug'25

- Published a paper for industrial efficiency enhancement using Computer Vision.
- Designed a chatbot agent with RAG along with agentic AI and MCP to streamline product support, reducing query resolution time.
- Developed product to use LLMs to auto extract insights from various databases including graphDB.
- Automated anomaly detection in industrial safety use cases with ML models & computer vision.
- Deployed AI models using Docker & APIs, ensuring scalable solutions for regulatory and operational insights.

Technical Intern, SIEMENS, Bengaluru

Feb'23-Jun'23

- Built a predictive model using linear regression for EV car price forecasting.
- Developed a React-Native dashboard app integrating OEE (Overall Equipment Efficiency) for manufacturing analytics, aligning with creative data visualization insights.

Research Intern, SIEMENS, Bengaluru

Aug'22-Oct'22

- Developed streamlined motor control techniques through the implementation of PWM (Pulse Width Modulation) signals, enhancing DC motor operational efficiency and performance
- Created drivers for Texas Instruments and Microchip microcontroller units

PUBLICATIONS

- Arya Darshan Roi Vijai, Lohith MV, Raghunandan KS, Kaundinya SR, 'Quality Check Efficiency Enhancement Through Accelerated Surface Analysis' [published](#) on Priorartregister.com, 2025, Disclosure Nr: OTT012971
- Arya Darshan Roi Vijai, Vanamala HR, Aayush Subramaniam, Aditya Rao, Akshat Vajrapu, 'Real Time Monitoring of Forest Fires and Wildfire Spread Prediction' [published](#) in IEEE Xplore Digital Library, 2023, pp. 772-779

ACADEMIC PROJECTS

Title: User Engagement Patterns on E-commerce Platforms

Team size: 1

Summary: This project utilizes the Pandas library to visualize diverse consumer behaviors on e-commerce sites and discern product viewing intentions. It applies K-means clustering and SVM modeling for analysis to provide insights into user engagement.

Title: Real Time Monitoring of Forest Fires and Wildfire Spread Prediction

Team size: 4

Summary: This project involved detecting forest fires and the direction and rate of spread of the fires using satellites and sensors for real-time detection and tracking, aiding quick response and predicting spread.

Title: Electronic Health Record System Using Blockchain

Team size: 4

Summary: This project aims to digitize health records using a decentralized system for enhanced data security. Created a secure patient health record database in blockchain with safe cryptographic techniques like block cipher.

Title: Joint Radar Communication

Team Size: 4

Summary: The potential of radar and communication on a single platform was employed in vehicle-to-vehicle communication in autonomous cars, enabling lower interferences between the two.

AWARDS & CERTIFICATIONS

- IP Samman award for early publication, SIEMENS, Sep'24
- SIEMENS AI Hackathon Finalist, Dec'23
- IBM AI Engineering Professional Certification, Coursera, Jan'24
- Earned the Distinction Award for Engineering Academic Excellence, PES University, Aug'20

PRESENTATIONS AND WORKSHOPS

- PowerPoint presentation on chatbot projects during the 'MLXPO Series 1 & 2' at SIEMENS, Bangalore, 2024
- Participated in 'AWS DeepRacer' conducted by Amazon, held at SIEMENS, Bengaluru, Nov'23
- Participated in and presented a paper on 'Real Time Monitoring of Forest Fires and Wildfire Spread Prediction' at the 2nd International Conference on Applied Artificial Intelligence and Computing (ICAAIC), Salem, India, 2023
- Completed two terms with the Centre for Innovation and Entrepreneurship, attending workshops on design thinking, agile practices, and business development, PES University, Aug'20 and Jan'21

TECHNICAL SKILLS

- Programming Languages: Python, JavaScript, Java
- Visualization: Tableau & Python
- ML/AI: Predictive Modeling, GenAI, NLP.
- Big Data & Cloud: AWS S3, Docker, FastAPI
- Tools: GitLab, Streamlit, React-native

EXTRACURRICULAR ACTIVITIES

- Organised the MLXPO (Machine Learning Expo) event at SIEMENS, Bangalore, 07'24 & 09'24
- Participated in the SIEMENS Cricket and Table Tennis Tournaments, held in Bangalore, 2024
- Co-organized and coordinated a Science Exhibition for high school kids at PES University, Oct'19
- Volunteered as a tutor for visually impaired children at Shree Ramana Maharishi Academy for the Blind, Bangalore, Jun'18